Abhinav Karnatak

Software Engineer | Full Stack & Applied AI

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Software engineer building **full-stack applications** and **AI agents** using TypeScript, Python, React, Node.js, and LangGraph. Built and deployed products involving **secure code execution, GitHub automation, real-time communication, payments, and cloud infrastructure.** I enjoy taking ownership of problems and building practical systems from idea to deployment.

SKILLS

AI Engineering: Agent Workflows, LangGraph, LangChain, RAG, Tool Calling, LLM API Integration, Prompt Engineering, AI Observability
Languages: Python, TypeScript, JavaScript, C++
Backend & Integrations: Node.js**,** Express.js, REST APIs, GitHub Apps, OAuth 2.0, JWT, E2B, WebSockets, Socket.io, RabbitMQ, Redis, Razorpay Payment Gateway
Databases: MongoDB, PostgreSQL, Drizzle ORM, Prisma
Cloud & DevOps: Docker, AWS EC2, AWS S3, Cloudflare R2, Nginx
Frontend Technologies: React.js, Next.js, React Native, Tailwind CSS, Framer Motion
Observability: PostHog

PROJECTS

Nimbus: Cloud Development Agent

- Built a **cloud development agent** that connects **GitHub repositories**, answers **codebase questions**, researches **the web**, and turns **natural-language requests into verified software changes**.
- Enabled **file inspection, code changes, command execution, file/artifact generation, Mermaid diagrams, and verification** inside isolated sandboxes**.**
- Added **pause-and-resume sessions**, follow-up execution, skills, attachments, and automated GitHub PR creation, updates, merges, and closures.

WakeOps: Automated Incident Response Platform

- Built a full-stack **incident response platform** using **Next.js, Node.js, PostgreSQL, and Prisma** that **converts Grafana alerts into tracked incidents** mapped to the correct service and on-call engineer.
- Created a **Temporal workflow** that places **automated Twilio calls, retries unanswered calls, escalates to a senior engineer**, and records acknowledgement and call history.
- Integrated RabbitMQ-based email and Slack notifications so alerts are delivered independently **without delaying the call escalation flow.**

YapIt: Real-Time Microservices Chat Application

- Built a **real-time chat application** using Next.js, Node.js, Socket.io, and MongoDB, with OTP-based authentication and WebSocket communication.
- Split the backend into **User, Chat, and Mail microservices**, using RabbitMQ for **asynchronous email delivery and account-cleanup tasks**.
- Containerized the services using Docker Compose and **deployed them on AWS EC2 behind Nginx**, with images stored in AWS S3 and OTPs temporarily stored in Redis.

Revuea: Empowering Honest Feedback with Anonymity

- Built a **full-stack anonymous feedback platform** using React, Node.js, PostgreSQL, and Prisma.
- Implemented **OTP-based authentication** and an analytics dashboard for visualizing feedback data.
- Integrated Gemini to **summarize large sets of feedback** and added CSV export for external analysis.

WORK EXPERIENCE

Analyst Sep 2025 - May 2026
Capgemini *Mumbai, Maharashtra*

- Resolved network and connectivity incidents across enterprise Linux and Windows environments while managing assigned support tickets within operational timelines.
- Monitored system health, investigated recurring issues, and coordinated with relevant teams when incidents required escalation.

Data Engineer Intern Jun 2024 - Jul 2024
HERE Technologies *Remote*

- Built custom Python web crawlers using **Scrapy** to scrape store locations from retail websites with varying HTML layouts, scheduling automated runs that cut manual effort by 30%.
- Documented the workflow and codebase to make it easier to maintain and update in the future.

EDUCATION

B.Tech Computer Science and Engineering Aug 2021 - May 2025
University Of Petroleum and Energy Studies

- Served as the **Student Placement Representative (SPR)**, coordinating communication between the placement cell and students during campus recruitment activities.